Multivariate Data Analysis
Lab Assignment 1

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Reg no : 24MSD7035

2025-08-05

```r
## QUESTION 1. Generate the multivariate data from R and test the goodness of fit. Represent the generated data with suitable graphs."

library(MASS)

## Warning: package 'MASS' was built under R version 4.4.3

library(MVN)

## Warning: package 'MVN' was built under R version 4.4.3

library(corrplot)

## Warning: package 'corrplot' was built under R version 4.4.3

## corrplot 0.95 loaded

# Mean vector and covariance matrix
mmu <- c(5, 10, 15)
mmd<- matrix(c(4, 2, 1,
                       2, 5, 3,
                       1, 3, 6), nrow = 3)

# Generate data
set.seed(982)
multi_data <- mvrnorm(n = 200, mmu, mmd)

#
mmu <- c(5, 10, 15)
mmd<- matrix(c(4, 2, 1,
                       2, 5, 3,
                       1, 3, 6), nrow = 3)

# Generate data
set.seed(92)
multi_data <- mvrnorm(n = 200, mmu, mmd)
multi_data <- as.data.frame(multi_data)
colnames(multi_data) <- c("X1", "X2", "X3")

result <- mvn(data = multi_data, mvn_test = "hz")
result
```

```
## $multivariate_normality
##            Test Statistic p.value     Method      MVN
## 1 Henze-Zirkler     0.803   0.386 asymptotic ✓ Normal
##
## $univariate_normality
##               Test Variable Statistic p.value Normality
## 1 Anderson-Darling       X1     0.388   0.384  ✓ Normal
## 2 Anderson-Darling       X2     0.152   0.960  ✓ Normal
## 3 Anderson-Darling       X3     0.345   0.483  ✓ Normal
##
## $descriptives
##   Variable   n   Mean Std.Dev Median    Min    Max   25th   75th  Skew
Kurtosis
## 1       X1 200  5.111   2.072  5.106 -1.858 11.429  3.668  6.654 0.027
2.931
## 2       X2 200 10.250   2.344 10.210  3.745 16.783  8.595 11.759 0.024
2.845
## 3       X3 200 15.193   2.351 15.156  9.906 22.521 13.363 16.797 0.182
2.788
##
## $data
##            X1        X2        X3
## 1    4.189759 13.915299 18.422168
## 2    4.755246  9.925191 17.440728
## 3    6.673078  8.987328 10.862712
## 4    6.099255 12.211415 17.684969
## 5    7.599408 13.966180 17.741801
## 6    6.271539 11.404812 17.418135
## 7    3.672074  6.318897 12.779381
## 8    4.630859 10.448097 15.188917
## 9    8.215495 11.477234 16.750621
## 10   4.523656 10.933534 15.568198
## 11   5.267675 10.769685 16.502790
## 12   7.716939 10.522629 13.966533
## 13   2.794622  5.045968  9.905864
## 14   3.539642 12.997925 16.934929
## 15   3.990326  8.648745 12.326435
## 16   8.202292 10.909712 17.488872
## 17   7.210545 10.153316 13.232988
## 18   7.090030 14.460625 16.794887
## 19   4.615623  9.485474 13.847524
## 20   4.907425 13.086548 14.678300
## 21   4.141334  7.116766 13.476976
## 22   5.142843  9.729761 14.326900
## 23   4.739552 10.630939 16.940027
## 24   2.786351  8.101997 11.781472
## 25   4.362308  6.868485 13.492445
## 26   2.900015  9.458617 16.248545
## 27   7.136486  9.394294 14.825977
## 28   1.139820  6.698081 13.611828
```

```
## 29   4.833182 12.270443 20.986996
## 30   6.699857 11.088787 18.434252
## 31   8.540123 12.348268 16.940606
## 32   2.761139  7.525325 13.261198
## 33   5.628191 10.263476 15.369094
## 34   5.850897 12.644076 16.487782
## 35   3.291128 10.206512 15.214851
## 36   7.197030  9.739678 13.370673
## 37   1.995597  7.761253 15.565677
## 38   9.158307  9.577516 12.023145
## 39   6.089988  7.443168 14.199894
## 40   4.349736 10.146218 16.165229
## 41   6.008411  9.448309 12.805847
## 42   3.785134 11.805210 12.781600
## 43   3.928749 12.503869 11.305695
## 44   3.325251  8.032473 15.591191
## 45   4.561072 10.574703 17.134044
## 46   3.352063  6.316674 11.665299
## 47   2.277389 10.153483 14.539152
## 48   8.462996 11.667838 16.033675
## 49   5.181909 11.215482 12.506523
## 50   5.773285 10.309900 13.192108
## 51   5.952519  9.164637 16.908362
## 52   7.822958  7.817924 12.994980
## 53   3.266698  8.689373 12.347417
## 54   4.081456 10.212177 14.034260
## 55   4.908192 11.062164 14.814454
## 56   7.295206 12.544397 14.069019
## 57   1.678386  6.132562 14.985422
## 58   4.843240 13.970458 16.328566
## 59   5.547234  9.203973 12.770813
## 60   5.886218 12.469982 17.174420
## 61   7.015751 10.852846 18.111014
## 62   3.883243  8.283561 15.447513
## 63   5.675019  9.844689 12.667427
## 64   6.998983 12.594484 17.406436
## 65   6.171909 12.511131 18.403070
## 66   5.835674 16.782958 18.498486
## 67   5.790307  8.407654 15.337503
## 68   4.454107  7.548801 13.302472
## 69   3.709472  5.073682 12.020904
## 70   2.959545  8.094373 13.270125
## 71   8.058031 10.774711 15.091495
## 72  -1.857507  5.299749 11.708781
## 73   3.891227 12.491953 15.958727
## 74   5.701542  9.869397 13.095409
## 75   3.921575 10.386782 13.984109
## 76   2.911228  9.234456 13.519462
## 77   4.278583  8.760101 17.156183
## 78   5.636386 11.871902 16.100488
```

```
## 79   7.632446  9.122680 15.801720
## 80   6.881870 10.388122 18.394993
## 81   4.345036  9.812664 15.844580
## 82   4.869409  9.489020 14.600885
## 83   5.771602 10.125258 13.749510
## 84   6.055307  9.387450 15.879034
## 85   3.631452  5.610141 15.286410
## 86   3.657645  7.486209 14.416884
## 87   6.124709 11.860565 10.349529
## 88   1.874743  7.434582 13.402629
## 89   4.446018  8.579899 15.273625
## 90   2.217595  7.937711 14.002780
## 91   1.738253  8.647097 15.443745
## 92   7.270112  8.077946 15.244031
## 93   7.823256 13.390507 17.298424
## 94   4.433342 10.599507 14.361977
## 95   4.296710 11.141157 17.415967
## 96   4.323854  8.297489 12.824100
## 97   6.793931 12.123817 16.732611
## 98   5.484060 10.207892 13.044738
## 99   5.669452  7.387264 13.041559
## 100  3.513225  9.363273 15.082151
## 101  5.882510 11.711486 18.267382
## 102  1.788061  5.592375 13.332585
## 103  4.721943  7.626074 13.686341
## 104  2.724990  3.745141 13.068239
## 105  5.913110  9.041526 12.437777
## 106  5.412591  8.817123 11.638215
## 107  4.952073  8.565972 11.867019
## 108  5.751373  9.360195 10.714357
## 109  2.934309 11.171687 17.805538
## 110  6.360805 15.040844 19.838766
## 111  7.503845 12.946093 16.688856
## 112  4.399598  8.933704 13.990490
## 113  6.273384  9.649209 16.546038
## 114  1.747473  8.414191 15.040775
## 115  3.407321 11.639269 17.636808
## 116  2.041123  8.194397 10.989308
## 117  6.093968  7.318305 13.100473
## 118  2.702875 11.419138 15.605530
## 119  3.125583  7.663835 16.202668
## 120  5.007178 12.295497 15.991073
## 121  7.035803  7.740619 14.247957
## 122  3.711799  8.014418 16.342448
## 123  4.821947 14.771680 18.590813
## 124  7.502776 13.279682 16.455470
## 125  6.366282  9.787060 17.737372
## 126  8.252696 12.966298 16.855912
## 127  7.203408 13.380664 19.833810
## 128  6.239941 11.042298 15.807147
```

```
## 129  4.151724  9.488078 12.455907
## 130  5.571163 10.924395 15.772843
## 131  3.690587 11.233110 16.621079
## 132 10.021361 13.407666 17.934518
## 133  1.665483  7.214102 13.167309
## 134  6.420304 10.735177 14.508920
## 135  7.687606 13.472666 15.123580
## 136  4.569545  8.667600 14.265592
## 137  2.591379  9.000908 16.240523
## 138  8.613028  8.706319 15.302568
## 139  5.878652 10.116693 12.041103
## 140  6.647709  5.189713 12.092312
## 141  7.411114 10.977345 16.505059
## 142 11.428911 13.988507 14.693095
## 143  7.606207 11.847125 18.301056
## 144  2.147960 11.019083 15.201239
## 145  4.677802 10.378889 13.796243
## 146  3.446876 12.768474 18.381071
## 147  6.770574 13.253781 16.740020
## 148  8.416372 13.532228 17.859867
## 149  7.149714 10.472397 12.253455
## 150  4.502473  9.955694 16.787392
## 151  4.286252 10.718397 10.053116
## 152  4.477795 10.247496 14.520199
## 153  5.841471 11.711836 15.038017
## 154  6.325754  9.973769 18.698401
## 155  1.611042 10.490663 14.219615
## 156  5.477307  8.392582 11.448554
## 157  5.375986 12.682834 16.802804
## 158  5.175573 12.570343 21.744706
## 159  5.364754 14.742819 18.801795
## 160  8.675535 13.637512 16.096044
## 161  2.503314  7.475816 11.440246
## 162  4.381179  8.040381 17.779759
## 163  5.291970 12.845475 16.722145
## 164  2.762378  8.599641 14.059561
## 165  5.560193  9.187193 12.808925
## 166  7.998240  9.387030 15.123896
## 167  1.816578  8.427253 15.298796
## 168  2.558271 13.565459 17.940429
## 169  2.339102 12.354786 18.446221
## 170  3.595676 11.009879 19.880605
## 171  3.264430  9.382406 10.716017
## 172  3.679274  8.327616 13.469649
## 173  6.852828 16.523566 22.521122
## 174  7.315189 13.211954 14.706852
## 175  7.768380 11.127261 15.340815
## 176  2.105857  6.309081 14.519421
## 177  5.423810 11.393788 14.678693
## 178  2.429776  9.347822 16.601024
```

```
## 179  2.236416  7.008256 14.098823
## 180  3.725294 11.662874 17.286451
## 181  1.606913  7.561981 11.548041
## 182  3.249456  9.997391 13.338175
## 183  4.266534 13.246208 14.652906
## 184  6.325683 11.347178 14.476446
## 185  7.217803 11.703403 19.250999
## 186  8.484917 11.472395 12.965814
## 187  8.524760 11.487307 16.725566
## 188  1.989808  6.936302 12.335004
## 189  1.584909  8.177773 17.731100
## 190  5.583360 13.419003 18.888430
## 191  5.477907 11.743309 18.942585
## 192  2.430807  9.740963 16.446214
## 193  4.207353 14.643113 16.719461
## 194  9.935768 15.033241 18.699927
## 195  7.236801  9.454752 12.891025
## 196  6.257910  7.865503 14.167647
## 197  7.213015 11.594152 14.762839
## 198  8.677508 13.032682 17.298789
## 199  5.068646 10.993764 14.568527
## 200  7.109312 10.669814 14.214801
##
## $subset
## NULL
##
## $outlierMethod
## [1] "none"
##
## attr(,"class")
## [1] "mvn"
```

```
result_mardia <- mvn(data = multi_data, mvn_test = "mardia")
result
```

```
## $multivariate_normality
##            Test Statistic p.value     Method      MVN
## 1 Henze-Zirkler     0.803   0.386 asymptotic ✓ Normal
##
## $univariate_normality
##               Test Variable Statistic p.value Normality
## 1 Anderson-Darling       X1     0.388   0.384  ✓ Normal
## 2 Anderson-Darling       X2     0.152   0.960  ✓ Normal
## 3 Anderson-Darling       X3     0.345   0.483  ✓ Normal
##
## $descriptives
##   Variable   n   Mean Std.Dev Median    Min    Max   25th   75th  Skew
Kurtosis
## 1       X1 200  5.111   2.072  5.106 -1.858 11.429  3.668  6.654 0.027
2.931
```

```
## 2       X2 200 10.250   2.344 10.210  3.745 16.783  8.595 11.759 0.024
2.845
## 3       X3 200 15.193   2.351 15.156  9.906 22.521 13.363 16.797 0.182
2.788
##
## $data
##            X1        X2        X3
## 1    4.189759 13.915299 18.422168
## 2    4.755246  9.925191 17.440728
## 3    6.673078  8.987328 10.862712
## 4    6.099255 12.211415 17.684969
## 5    7.599408 13.966180 17.741801
## 6    6.271539 11.404812 17.418135
## 7    3.672074  6.318897 12.779381
## 8    4.630859 10.448097 15.188917
## 9    8.215495 11.477234 16.750621
## 10   4.523656 10.933534 15.568198
## 11   5.267675 10.769685 16.502790
## 12   7.716939 10.522629 13.966533
## 13   2.794622  5.045968  9.905864
## 14   3.539642 12.997925 16.934929
## 15   3.990326  8.648745 12.326435
## 16   8.202292 10.909712 17.488872
## 17   7.210545 10.153316 13.232988
## 18   7.090030 14.460625 16.794887
## 19   4.615623  9.485474 13.847524
## 20   4.907425 13.086548 14.678300
## 21   4.141334  7.116766 13.476976
## 22   5.142843  9.729761 14.326900
## 23   4.739552 10.630939 16.940027
## 24   2.786351  8.101997 11.781472
## 25   4.362308  6.868485 13.492445
## 26   2.900015  9.458617 16.248545
## 27   7.136486  9.394294 14.825977
## 28   1.139820  6.698081 13.611828
## 29   4.833182 12.270443 20.986996
## 30   6.699857 11.088787 18.434252
## 31   8.540123 12.348268 16.940606
## 32   2.761139  7.525325 13.261198
## 33   5.628191 10.263476 15.369094
## 34   5.850897 12.644076 16.487782
## 35   3.291128 10.206512 15.214851
## 36   7.197030  9.739678 13.370673
## 37   1.995597  7.761253 15.565677
## 38   9.158307  9.577516 12.023145
## 39   6.089988  7.443168 14.199894
## 40   4.349736 10.146218 16.165229
## 41   6.008411  9.448309 12.805847
## 42   3.785134 11.805210 12.781600
## 43   3.928749 12.503869 11.305695
```

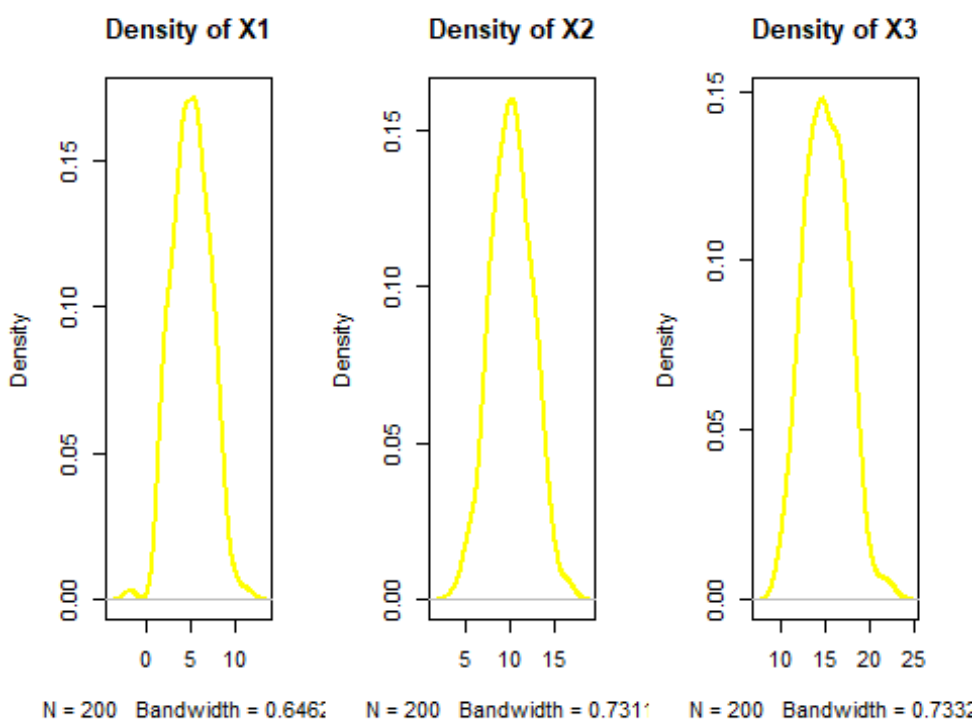
```
## 44   3.325251  8.032473 15.591191
## 45   4.561072 10.574703 17.134044
## 46   3.352063  6.316674 11.665299
## 47   2.277389 10.153483 14.539152
## 48   8.462996 11.667838 16.033675
## 49   5.181909 11.215482 12.506523
## 50   5.773285 10.309900 13.192108
## 51   5.952519  9.164637 16.908362
## 52   7.822958  7.817924 12.994980
## 53   3.266698  8.689373 12.347417
## 54   4.081456 10.212177 14.034260
## 55   4.908192 11.062164 14.814454
## 56   7.295206 12.544397 14.069019
## 57   1.678386  6.132562 14.985422
## 58   4.843240 13.970458 16.328566
## 59   5.547234  9.203973 12.770813
## 60   5.886218 12.469982 17.174420
## 61   7.015751 10.852846 18.111014
## 62   3.883243  8.283561 15.447513
## 63   5.675019  9.844689 12.667427
## 64   6.998983 12.594484 17.406436
## 65   6.171909 12.511131 18.403070
## 66   5.835674 16.782958 18.498486
## 67   5.790307  8.407654 15.337503
## 68   4.454107  7.548801 13.302472
## 69   3.709472  5.073682 12.020904
## 70   2.959545  8.094373 13.270125
## 71   8.058031 10.774711 15.091495
## 72  -1.857507  5.299749 11.708781
## 73   3.891227 12.491953 15.958727
## 74   5.701542  9.869397 13.095409
## 75   3.921575 10.386782 13.984109
## 76   2.911228  9.234456 13.519462
## 77   4.278583  8.760101 17.156183
## 78   5.636386 11.871902 16.100488
## 79   7.632446  9.122680 15.801720
## 80   6.881870 10.388122 18.394993
## 81   4.345036  9.812664 15.844580
## 82   4.869409  9.489020 14.600885
## 83   5.771602 10.125258 13.749510
## 84   6.055307  9.387450 15.879034
## 85   3.631452  5.610141 15.286410
## 86   3.657645  7.486209 14.416884
## 87   6.124709 11.860565 10.349529
## 88   1.874743  7.434582 13.402629
## 89   4.446018  8.579899 15.273625
## 90   2.217595  7.937711 14.002780
## 91   1.738253  8.647097 15.443745
## 92   7.270112  8.077946 15.244031
## 93   7.823256 13.390507 17.298424
```

```
## 94   4.433342 10.599507 14.361977
## 95   4.296710 11.141157 17.415967
## 96   4.323854  8.297489 12.824100
## 97   6.793931 12.123817 16.732611
## 98   5.484060 10.207892 13.044738
## 99   5.669452  7.387264 13.041559
## 100  3.513225  9.363273 15.082151
## 101  5.882510 11.711486 18.267382
## 102  1.788061  5.592375 13.332585
## 103  4.721943  7.626074 13.686341
## 104  2.724990  3.745141 13.068239
## 105  5.913110  9.041526 12.437777
## 106  5.412591  8.817123 11.638215
## 107  4.952073  8.565972 11.867019
## 108  5.751373  9.360195 10.714357
## 109  2.934309 11.171687 17.805538
## 110  6.360805 15.040844 19.838766
## 111  7.503845 12.946093 16.688856
## 112  4.399598  8.933704 13.990490
## 113  6.273384  9.649209 16.546038
## 114  1.747473  8.414191 15.040775
## 115  3.407321 11.639269 17.636808
## 116  2.041123  8.194397 10.989308
## 117  6.093968  7.318305 13.100473
## 118  2.702875 11.419138 15.605530
## 119  3.125583  7.663835 16.202668
## 120  5.007178 12.295497 15.991073
## 121  7.035803  7.740619 14.247957
## 122  3.711799  8.014418 16.342448
## 123  4.821947 14.771680 18.590813
## 124  7.502776 13.279682 16.455470
## 125  6.366282  9.787060 17.737372
## 126  8.252696 12.966298 16.855912
## 127  7.203408 13.380664 19.833810
## 128  6.239941 11.042298 15.807147
## 129  4.151724  9.488078 12.455907
## 130  5.571163 10.924395 15.772843
## 131  3.690587 11.233110 16.621079
## 132 10.021361 13.407666 17.934518
## 133  1.665483  7.214102 13.167309
## 134  6.420304 10.735177 14.508920
## 135  7.687606 13.472666 15.123580
## 136  4.569545  8.667600 14.265592
## 137  2.591379  9.000908 16.240523
## 138  8.613028  8.706319 15.302568
## 139  5.878652 10.116693 12.041103
## 140  6.647709  5.189713 12.092312
## 141  7.411114 10.977345 16.505059
## 142 11.428911 13.988507 14.693095
## 143  7.606207 11.847125 18.301056
```

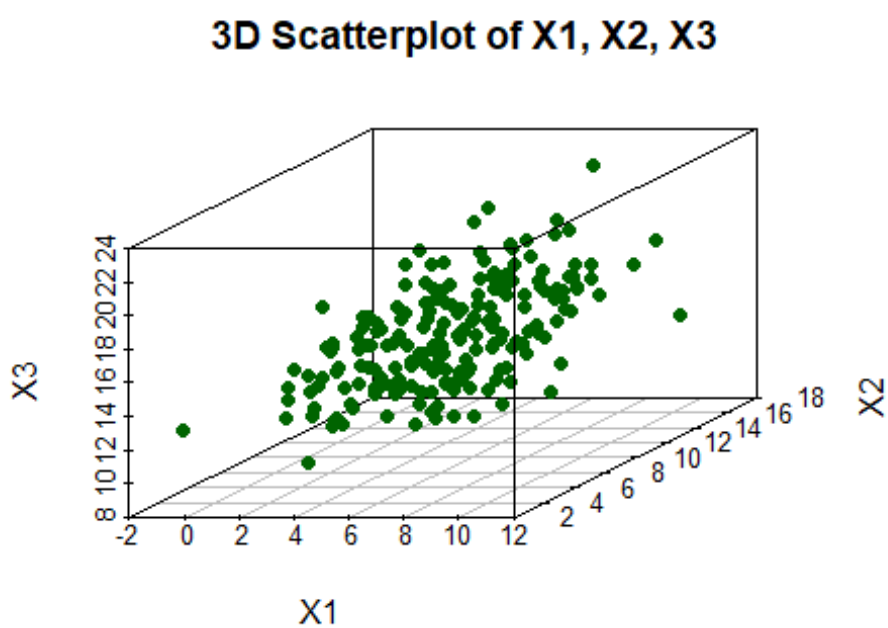
```
## 144  2.147960 11.019083 15.201239
## 145  4.677802 10.378889 13.796243
## 146  3.446876 12.768474 18.381071
## 147  6.770574 13.253781 16.740020
## 148  8.416372 13.532228 17.859867
## 149  7.149714 10.472397 12.253455
## 150  4.502473  9.955694 16.787392
## 151  4.286252 10.718397 10.053116
## 152  4.477795 10.247496 14.520199
## 153  5.841471 11.711836 15.038017
## 154  6.325754  9.973769 18.698401
## 155  1.611042 10.490663 14.219615
## 156  5.477307  8.392582 11.448554
## 157  5.375986 12.682834 16.802804
## 158  5.175573 12.570343 21.744706
## 159  5.364754 14.742819 18.801795
## 160  8.675535 13.637512 16.096044
## 161  2.503314  7.475816 11.440246
## 162  4.381179  8.040381 17.779759
## 163  5.291970 12.845475 16.722145
## 164  2.762378  8.599641 14.059561
## 165  5.560193  9.187193 12.808925
## 166  7.998240  9.387030 15.123896
## 167  1.816578  8.427253 15.298796
## 168  2.558271 13.565459 17.940429
## 169  2.339102 12.354786 18.446221
## 170  3.595676 11.009879 19.880605
## 171  3.264430  9.382406 10.716017
## 172  3.679274  8.327616 13.469649
## 173  6.852828 16.523566 22.521122
## 174  7.315189 13.211954 14.706852
## 175  7.768380 11.127261 15.340815
## 176  2.105857  6.309081 14.519421
## 177  5.423810 11.393788 14.678693
## 178  2.429776  9.347822 16.601024
## 179  2.236416  7.008256 14.098823
## 180  3.725294 11.662874 17.286451
## 181  1.606913  7.561981 11.548041
## 182  3.249456  9.997391 13.338175
## 183  4.266534 13.246208 14.652906
## 184  6.325683 11.347178 14.476446
## 185  7.217803 11.703403 19.250999
## 186  8.484917 11.472395 12.965814
## 187  8.524760 11.487307 16.725566
## 188  1.989808  6.936302 12.335004
## 189  1.584909  8.177773 17.731100
## 190  5.583360 13.419003 18.888430
## 191  5.477907 11.743309 18.942585
## 192  2.430807  9.740963 16.446214
## 193  4.207353 14.643113 16.719461
```

```
## 194  9.935768 15.033241 18.699927
## 195  7.236801  9.454752 12.891025
## 196  6.257910  7.865503 14.167647
## 197  7.213015 11.594152 14.762839
## 198  8.677508 13.032682 17.298789
## 199  5.068646 10.993764 14.568527
## 200  7.109312 10.669814 14.214801
##
## $subset
## NULL
##
## $outlierMethod
## [1] "none"
##
## attr(,"class")
## [1] "mvn"
```

```r
# Density Plot for Each Variable

par(mfrow = c(1, 3))
for (i in 1:3) {
  plot(density(multi_data[[i]]),
       main = paste("Density of", colnames(multi_data)[i]),
       col = "yellow", lwd = 2)
}
```

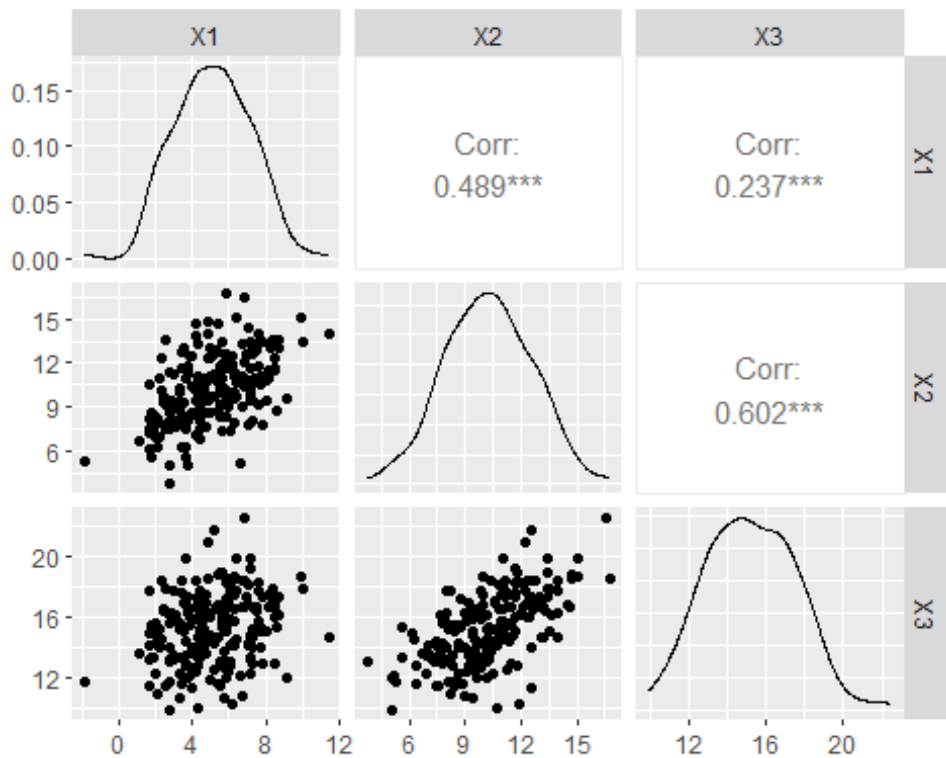


```r
library(scatterplot3d)

scatterplot3d(
  x = multi_data$X1,
  y = multi_data$X2,
  z = multi_data$X3,
  pch = 19,
  color = "darkgreen",
  main = "3D Scatterplot of X1, X2, X3",
  xlab = "X1",
  ylab = "X2",
  zlab = "X3"
)
```




```r
# Pairwise Density Plots (using GGally)


library(GGally)

## Warning: package 'GGally' was built under R version 4.4.3

## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.4.3

# Rename columns (if not done already)
colnames(multi_data) <- c("X1", "X2", "X3")
```

```r
# GGally plot
ggpairs(multi_data)
```




```r
## QUESTION 2.Select any one dataset of your own and test its goodness of fit. Give interpretations through plots.

data("USArrests")
head(USArrests)

##            Murder Assault UrbanPop Rape
## Alabama      13.2     236       58 21.2
## Alaska       10.0     263       48 44.5
## Arizona       8.1     294       80 31.0
## Arkansas      8.8     190       50 19.5
## California    9.0     276       91 40.6
## Colorado      7.9     204       78 38.7

# Run Mardia's Test
library(MVN)
output1 <- mvn(data = USArrests, mvn_test = "mardia")
output1

## $multivariate_normality
##               Test Statistic p.value     Method          MVN
## 1 Mardia Skewness    38.624   0.007 asymptotic ✗ Not normal
## 2 Mardia Kurtosis     0.962   0.336 asymptotic     ✓ Normal
##
```

```
## $univariate_normality
##                Test Variable Statistic p.value Normality
## 1 Anderson-Darling   Murder     0.603   0.111  ✓ Normal
## 2 Anderson-Darling  Assault     0.726   0.055  ✓ Normal
## 3 Anderson-Darling UrbanPop     0.319   0.525  ✓ Normal
## 4 Anderson-Darling     Rape     0.676   0.073  ✓ Normal
##
## $descriptives
##   Variable  n    Mean Std.Dev Median  Min   Max    25th    75th   Skew
Kurtosis
## 1   Murder 50   7.788   4.356   7.25  0.8  17.4   4.075  11.250  0.382
2.135
## 2  Assault 50 170.760  83.338 159.00 45.0 337.0 109.000 249.000  0.227
1.931
## 3 UrbanPop 50  65.540  14.475  66.00 32.0  91.0  54.500  77.750 -0.219
2.216
## 4     Rape 50  21.232   9.366  20.10  7.3  46.0  15.075  26.175  0.777
3.202
##
## $data
##                Murder Assault UrbanPop Rape
## Alabama          13.2     236       58 21.2
## Alaska           10.0     263       48 44.5
## Arizona           8.1     294       80 31.0
## Arkansas          8.8     190       50 19.5
## California        9.0     276       91 40.6
## Colorado          7.9     204       78 38.7
## Connecticut       3.3     110       77 11.1
## Delaware          5.9     238       72 15.8
## Florida          15.4     335       80 31.9
## Georgia          17.4     211       60 25.8
## Hawaii            5.3      46       83 20.2
## Idaho             2.6     120       54 14.2
## Illinois         10.4     249       83 24.0
## Indiana           7.2     113       65 21.0
## Iowa              2.2      56       57 11.3
## Kansas            6.0     115       66 18.0
## Kentucky          9.7     109       52 16.3
## Louisiana        15.4     249       66 22.2
## Maine             2.1      83       51  7.8
## Maryland         11.3     300       67 27.8
## Massachusetts     4.4     149       85 16.3
## Michigan         12.1     255       74 35.1
## Minnesota         2.7      72       66 14.9
## Mississippi      16.1     259       44 17.1
## Missouri          9.0     178       70 28.2
## Montana           6.0     109       53 16.4
## Nebraska          4.3     102       62 16.5
## Nevada           12.2     252       81 46.0
## New Hampshire     2.1      57       56  9.5
```

```
## New Jersey        7.4     159       89 18.8
## New Mexico       11.4     285       70 32.1
## New York         11.1     254       86 26.1
## North Carolina   13.0     337       45 16.1
## North Dakota      0.8      45       44  7.3
## Ohio              7.3     120       75 21.4
## Oklahoma          6.6     151       68 20.0
## Oregon            4.9     159       67 29.3
## Pennsylvania      6.3     106       72 14.9
## Rhode Island      3.4     174       87  8.3
## South Carolina   14.4     279       48 22.5
## South Dakota      3.8      86       45 12.8
## Tennessee        13.2     188       59 26.9
## Texas            12.7     201       80 25.5
## Utah              3.2     120       80 22.9
## Vermont           2.2      48       32 11.2
## Virginia          8.5     156       63 20.7
## Washington        4.0     145       73 26.2
## West Virginia     5.7      81       39  9.3
## Wisconsin         2.6      53       66 10.8
## Wyoming           6.8     161       60 15.6
##
## $subset
## NULL
##
## $outlierMethod
## [1] "none"
##
## attr(,"class")
## [1] "mvn"
```

1. Univariate Normality Test Results (Anderson–Darling)

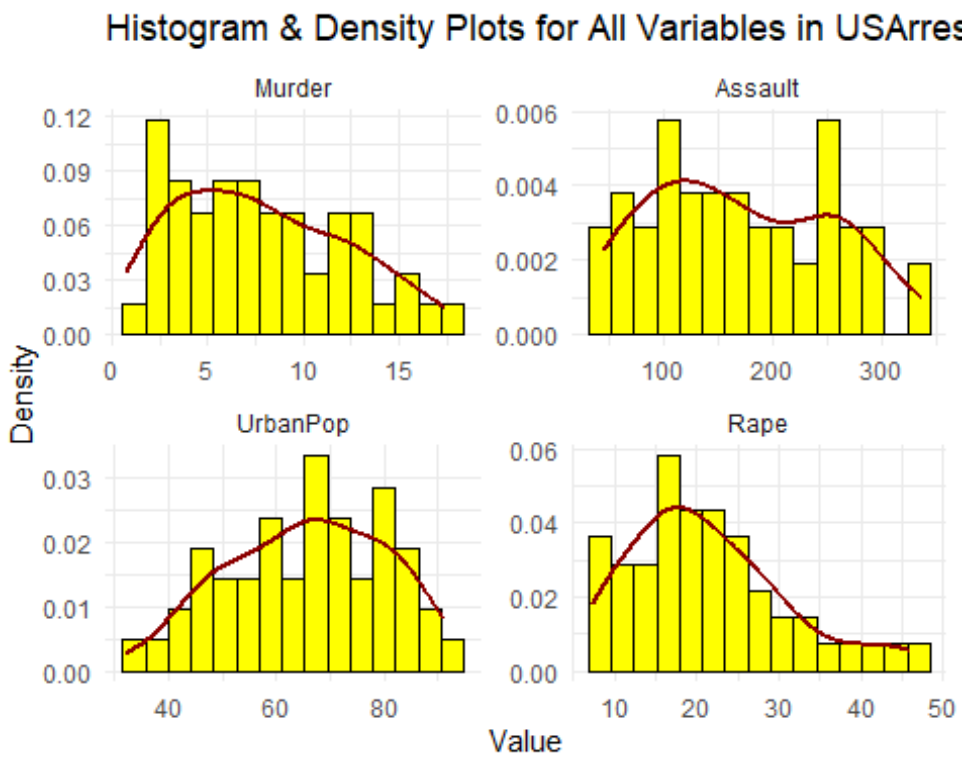
Interpretation:
All four variables passed the Anderson-Darling normality test (p > 0.05), indicating that each #variable has a normal distribution.

2. Multivariate Normality Test (Mardia's Test)

Component	Test Statistic	p-value	Result
Mardia Skewness	38.624	0.007	❌Not Normal
Mardia Kurtosis	0.962	0.336	✅Normal

Interpretation:
The data has adequate kurtosis (no heavy tails), but skewness is very high. This indicates that the dataset as a whole does not follow multivariate normality, even if individual variables do.

```r
Visual Plots for Interpretation


library(ggplot2)

# Convert to long format for faceted plotting
library(reshape2)

## Warning: package 'reshape2' was built under R version 4.4.3

data_long <- melt(USArrests)

## No id variables; using all as measure variables

ggplot(data_long, aes(x = value)) +
  geom_histogram(aes(y = after_stat(density)), bins = 15, fill = "yellow",
color = "black") +
  geom_density(color = "darkred", size = 1) +
  facet_wrap(~ variable, scales = "free", ncol = 2) +
  theme_minimal() +
  ggtitle("Histogram & Density Plots for All Variables in USArrests") +
  xlab("Value") +
  ylab("Density")

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
#This plot visualizes the distribution of four variables across U.S. states:

# Murder

The distribution is somewhat tilted to the right, indicating that while some
states have higher murder rates, the majority have low to moderate rates.

Outliers or irregular distributions are indicated by the density curve's
longer tail on the right.



# Assault

There is no obvious bell shape to the distribution, which seems to be multi-
modal or about uniform.

This implies that there is not an apparent central tendency in the assault
rates, which vary greatly between states.

Not typically distributed, for sure.

# UrbanPop

Although slightly flatter than a conventional Gaussian curve, the
distribution of the urban population appears to be quite symmetrical and
closer to normal.Shows that the proportion of people living in cities is
```

comparatively equal across states. This comes the closest to normalcy of the four.

 # Rape

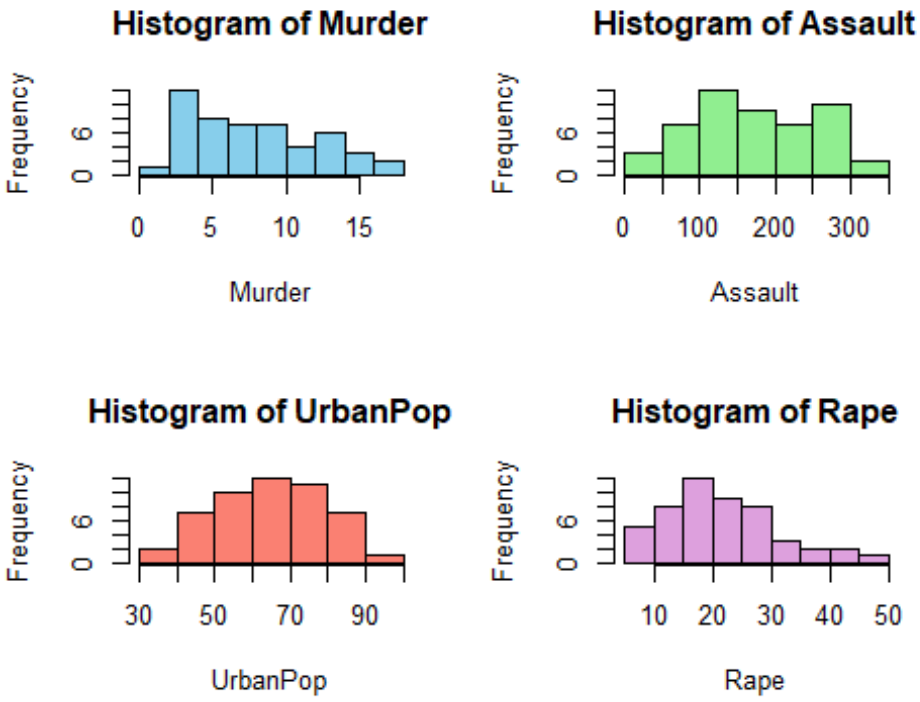
Like murder, the distribution is biased to the right.Rape rates are low to moderate in the majority of states, but they are noticeably higher in some.

The deviation from normality is probably confirmed by #skewness and kurtosis.

 # Final:

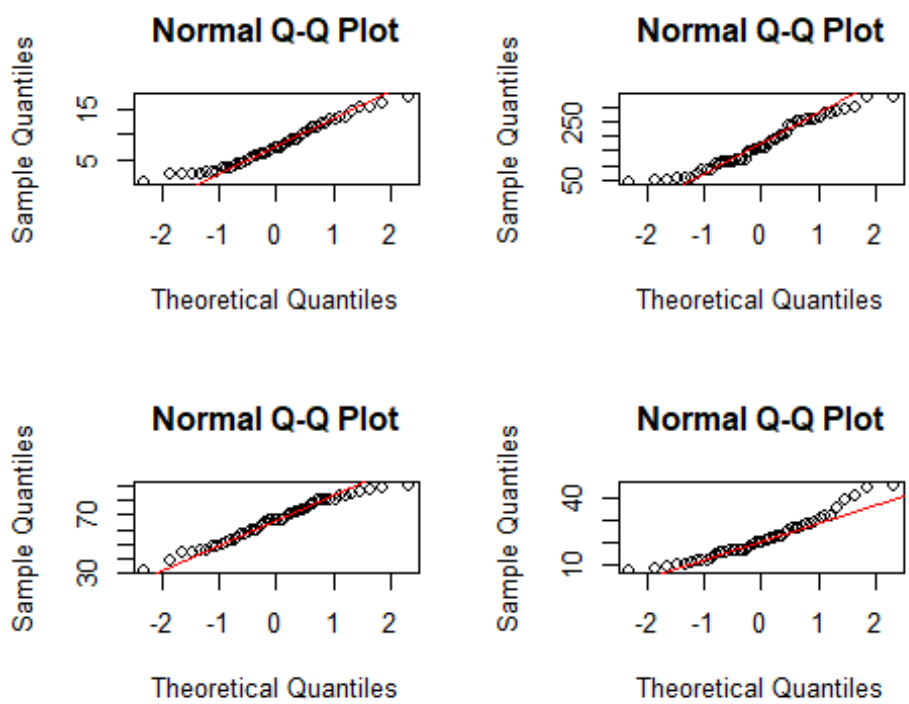
UrbanPop is the closest variable, but none of the others appear to be exactly normally distributed.

```r
# Normal histogram
par(mfrow = c(2, 2))

# Plot histograms for each variable
hist(USArrests$Murder,
     main = "Histogram of Murder",
     xlab = "Murder",
     col = "skyblue",
     border = "black")

hist(USArrests$Assault,
     main = "Histogram of Assault",
     xlab = "Assault",
     col = "lightgreen",
     border = "black")

hist(USArrests$UrbanPop,
     main = "Histogram of UrbanPop",
     xlab = "UrbanPop",
     col = "salmon",
     border = "black")

hist(USArrests$Rape,
     main = "Histogram of Rape",
     xlab = "Rape",
     col = "plum",
     border = "black")
```



1. Murder Histogram
Murder rates are low in the majority of states (0–5). Murder rates above 10 are found in very few states. Because of the data's right-skewedness, certain states have extremely high murder rates.

2. The Assault Histogram
The distribution of assault rates is more uniform. Assault rates range from 100 to 300 in many states.The data exhibits minimal skewness and a fairly homogeneous distribution.

3. The UrbanPop Histogram
Between 50 and 80 percent of people live in cities in the majority of states. The distribution is roughly bell-shaped or normal. Shows that the number of urban residents in each state is fairly equal.
4. Rape Histogram
Rape rates range from 10 to 30 in the majority of states.
A few states have greater rates, and the distribution is somewhat skewed to the right.
Implies that although the majority of states have moderate rates of rape, there are some exceptions.

```r
par(mfrow = c(2,2))
qqnorm(USArrests$Murder); qqline(USArrests$Murder, col = "red")
qqnorm(USArrests$Assault); qqline(USArrests$Assault, col = "red")
qqnorm(USArrests$UrbanPop); qqline(USArrests$UrbanPop, col = "red")
qqnorm(USArrests$Rape); qqline(USArrests$Rape, col = "red")
```